Budgeted Entanglement Routing for Sequential Multi-Agent Reinforcement Learning

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Abstract

Entanglement-assisted multi-agent learning usually treats entanglement as fixed or continuously available. We instead study a sequential resource problem in which four decentralized agents share six consumable Bell pairs over ten rounds of changing private-information coordination requests. Before private inputs arrive, a public controller must select a feasible matching of agent pairs to receive entanglement. We introduce BER-MARL, a masked Deep Q-Network that learns whether to spend or save Bell pairs from public request values, graph structure, budget, and horizon. Every request is a three-input binary-output XOR game. Its zero-communication classical value is obtained by exhaustive enumeration, while its Bell-assisted value is independently bounded by a semidefinite program. In a locked evaluation with 16 training seeds and 1,000 paired held-out sequences per seed, BER-MARL achieved mean entanglement surplus 0.5064, compared with 0.3084 for myopic greedy, 0.2268 for random, and 0.2248 for static allocation. The paired BER-MARL-minus-greedy difference was 0.1980, with 95% bootstrap interval $[0.1914, 0.2043]$; all seeds were positive. BER-MARL recovered 88.66% of the exact online dynamic-programming surplus. A preregistered condition requiring the one-sided lower bound to exceed 90% failed because the bound was 87.68%. The supported contribution is therefore a certified scarce-entanglement routing benchmark and robust learned allocation gain, rather than a near-oracle, hardware, or computational-advantage claim.

Keywords: quantum multi-agent reinforcement learning; entanglement allocation; nonlocal games; constrained reinforcement learning; dynamic matching.

1 Introduction

Entanglement can strengthen correlations between separated agents without revealing their private inputs [1, 2, 8]. Existing entanglement-assisted multi-agent learning work demonstrates that such correlations can be incorporated into learned decentralized policies [4]. A separate systems question arises when entanglement is scarce: if many agent pairs could benefit but only a few Bell pairs are available, which opportunities should receive them, and when should the resource be saved?

We formulate this as a finite-horizon resource-allocation problem. Four agents face a changing public graph of pairwise coordination requests. Each endpoint later receives a private input and returns one binary output. An allocator chooses a matching before those private inputs arrive. Selected edges use frozen Bell-assisted local policies and consume one Bell pair each; unselected edges use exact optimal zero-message classical policies. The controller begins with six pairs for ten rounds.

This decomposition separates two questions. At the *request level*, exhaustive classical enumeration and an XOR-game semidefinite program (SDP) certify how much one Bell pair can improve expected success. At the *sequence level*, reinforcement learning decides when and where to spend the certified resource. The resulting method, BER-MARL (Budgeted Entanglement Routing for MARL), uses a masked Deep Q-Network (DQN) [5] over feasible graph matchings.

Our contributions are:

1. a sequential benchmark for routing consumable entanglement over a dynamic matching-constrained multi-agent graph;
2. exact request-level classical values and independent SDP certification of Bell-assisted values;
3. BER-MARL, a public-information DQN allocator that learns budget pacing without private-context access;
4. a prospective held-out comparison against random, static, myopic greedy, exact online dynamic programming (DP), and privileged ceilings; and
5. complete raw results and an independent audit preserving both a strong learned-versus-greedy result and a failed near-oracle gate.

2 Related Work and Positioning

Bell nonlocality and quantum XOR games establish that entangled strategies can outperform local classical correlations under restricted communication [1, 2, 8, 3]. This paper does not propose a new Bell inequality. It uses heterogeneous small XOR games as exactly certifiable demand units.

Quantum multi-agent reinforcement learning has considered entangled value functions and policies in repeated coordination environments. In particular, eQMARL stud-

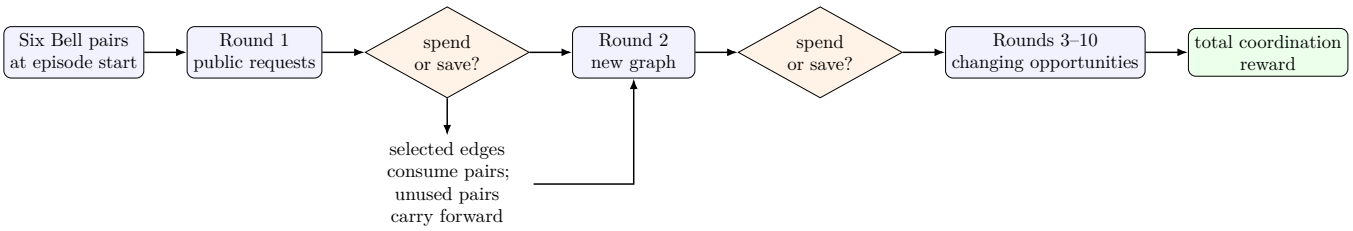


Figure 1: The sequential allocation problem. A myopic policy spends on the best visible requests. A long-horizon policy may preserve Bell pairs for rarer, higher-value future states.

ies distributed cooperation over quantum channels [4]. Our focus differs: local request policies are frozen before allocator training, Bell pairs are explicitly consumable, a graph matching limits simultaneous allocations, and the primary learned object is the long-horizon resource controller.

The allocator is related to constrained reinforcement learning and finite-horizon resource scheduling [7, 5]. DQN is used because the registered action set is discrete and small. PPO [6], recurrent policies, model-based RL, and combinatorial action architectures are important future baselines. A bounded novelty screen found no retrieved study combining consumable Bell-pair budgets, dynamic matching constraints, decentralized request rewards, and a learned long-horizon allocator. We therefore avoid an unqualified priority claim pending a systematic human-reviewed literature search.

3 Game Setting

3.1 One private-information request

Each endpoint privately observes $x, y \in \{0, 1, 2\}$ and emits one bit $a, b \in \{0, 1\}$. A public predicate $f : \{0, 1, 2\}^2 \to \{0, 1\}$ defines success:

$$a \oplus b = f(x, y). \tag{1}$$

The outputs are abstract coordination answers. They do not move a robot or select a physical channel in the implemented benchmark. Their purpose is to measure whether separated agents can satisfy a joint rule without sharing private inputs.

For example, if $f(1, 2) = 1$, both $(a, b) = (0, 1)$ and $(1, 0)$ win; neither agent is assigned a particular bit. If $f(1, 2) = 0$, both $(0, 0)$ and $(1, 1)$ win. A single input pair is trivial, but the agents must use one local rule over all nine input pairs. Conflicting cells can prevent perfect local play. For example, requiring "same" for $(0, 0), (0, 1), (1, 0)$ and "different" for $(1, 1)$ is inconsistent with fixed deterministic local outputs.

The input pair is sampled from a public, nonuniform positive weight matrix w_{xy}. For deterministic local functions g, h, expected success is

$$V(g, h) = \sum_{x,y} w_{xy} \, \mathbf{1}[g(x) \oplus h(y) = f(x, y)] \,. \tag{2}$$

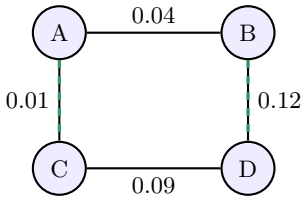


Figure 2: Example public request graph. Edge labels are certified immediate gains. The dashed matching $\{A\text{–}!C, B\text{–}!D\}$ is feasible; allocations sharing an endpoint are not. Immediate gain alone may still be inferior to saving budget.

There are eight binary functions of a three-valued input, hence only $8 \times 8 = 64$ deterministic pairs. We enumerate all of them:

$$V_C = \max_{g,h} V(g, h). \tag{3}$$

Shared classical randomness cannot exceed this optimum because expected reward is linear in a mixture of deterministic strategies.

3.2 Bell-assisted request policy

On a selected edge, endpoints share

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \tag{4}$$

Each endpoint chooses a local binary measurement from its own private input. Six measurement angles are optimized to obtain V_Q. An independent XOR-game SDP supplies an upper bound on the quantum bias [3, 8]; the optimized policies match the bound to registered tolerance.

With urgency u_e, the certified expected gain from spending one Bell pair on request e is

$$G_e = u_e \max(0, V_{Q,e} - V_{C,e}). \tag{5}$$

Zero-gap requests remain in the family, so entangling every active edge is not automatically useful.

3.3 Ten-round allocation problem

The environment has four agents, ten rounds, and an initial budget $B_1 = 6$. Three to five of the six possible edges are active per round. Each active edge carries a request type

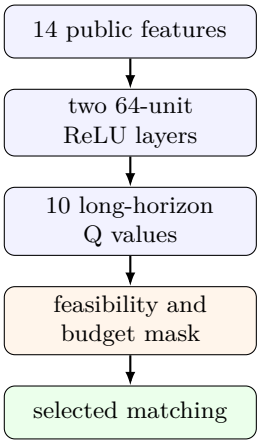


Figure 3: BER-MARL architecture. The network receives no private endpoint context.

and urgency. The public graph state follows a committed Markov transition matrix.

Before private inputs arrive, the allocator observes the public graph, gains, time, and budget. It selects a matching M_t of active edges satisfying $|M_t| \le B_t$. Selected edges use frozen Bell policies; other edges use frozen exact-classical policies. An agent may answer multiple requests classically but cannot receive entanglement on multiple edges in one round.

The immediate surplus and budget transition are

$$r_t = \sum_{e \in M_t} G_e, \tag{6}$$

$$B_{t+1} = B_t - |M_t|. \tag{7}$$

The primary outcome is $S = \sum_{t=1}^{10} r_t$, equal to total expected return minus exact classical return on the same public sequence. The implementation integrates over private-input distributions using certified V_C and V_Q; individual contexts are not sampled during allocator learning. This isolates scheduling from local-game sampling noise.

4 Proposed Method: BER-MARL

The state vector contains six current edge gains, six activity indicators, normalized budget, and normalized remaining horizon. A two-layer multilayer perceptron with 64 units per layer outputs Q values for ten global actions: the empty matching, six single edges, and three disjoint two-edge matchings. A deterministic mask removes inactive, endpoint-conflicting, and over-budget actions.

For transition (s_t, a_t, r_t, s_{t+1}), DQN minimizes a Huber loss between $Q_\theta(s_t, a_t)$ and

$$y_t = r_t + \max_{a' \in \mathcal{A}_{\text{feas}}(s_{t+1})} Q_{\bar{\theta}}(s_{t+1}, a'), \tag{8}$$

using a replay buffer and target parameters $\bar{\theta}$. Training uses Adam at 5×10^{-4}, batch size 64, target updates every 200 optimizer steps, and 2,500 episodes per final seed. The endpoint policies are certified and frozen before DQN training; BER-MARL learns only resource allocation.

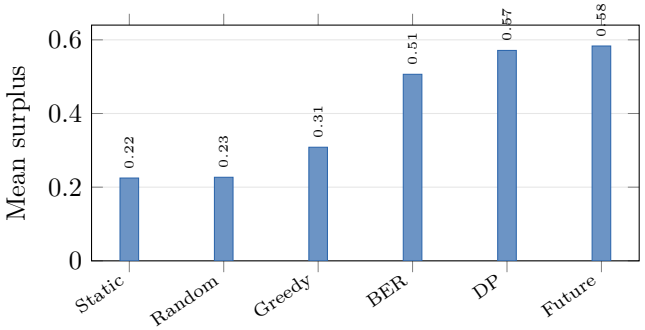


Figure 4: Held-out surplus. BER-MARL closes most of the gap from greedy to model-aware DP.

5 Benchmarking Methods

The DP value recursion is

$$J_t(s,b) = \max_{M \in \mathcal{M}(s,b)} \Big[G(s,M) + \sum_{s'} P(s'|s) J_{t+1}(s', b - |M|) \Big]. \tag{9}$$

DP and SDP play different roles: SDP certifies one request's quantum value, while DP provides an exact model-aware allocation reference over time.

6 Experimental Protocol

The study was executed prospectively in five stages: plan and claims; request/task certification; bounded algorithm development and freeze; locked held-out evaluation; and independent release audit. Development considered four DQN configurations with two seeds each, followed by four fresh pilots. Before final access, the protocol fixed 16 training seeds, 2,500 training episodes per seed, 1,000 paired test sequences, final-checkpoint evaluation, 20,000 seed-level bootstrap resamples, and five positive-result gates.

Final models train on development and validation catalogs. Evaluation uses disjoint request types, a disjoint public transition catalog, and a disjoint sequence stream without adaptation. The independent training seed is the unit of inference; episodes within a seed are not treated as independent replicates.

The preregistered composite gate required: (i) a paired 95% lower bound for BER-MARL-minus-greedy above zero; (ii) mean improvement at least 5% of DP surplus; (iii) a one-sided 95% lower bound for the BER-MARL/DP fraction above 0.90; (iv) a one-sided surplus lower bound above zero; and (v) no budget or matching violation.

7 Results

BER-MARL exceeded greedy by 0.19803. The paired-seed 95% bootstrap interval was $[0.19141, 0.20432]$, and all 16 seeds were positive. Relative to greedy surplus, the

Table 1: Registered comparison methods. Random, static, greedy, and BER-MARL have identical public information, local policies, sequences, matching constraints, and six-pair budget.

Method	Bell budget	Future/model access	Role
Classical exact	0	None	Defines zero surplus using exact optimal local strategies.
Quantum random	6	None	Tests whether arbitrary entanglement use is sufficient.
Quantum static	6	Non-test frequencies	Uses fixed edge priorities; does not respond to future value.
Quantum greedy	6	Current round	Selects the greatest visible certified gain; primary comparator.
BER-MARL	6	Learned experience	Proposed long-horizon public-information allocator.
Online DP	6	True transition matrix	Exact non-clairvoyant model-aware reference.
Future-clairvoyant	6	Realized future	Unattainable descriptive ceiling.
Message oracle	Not comparable	Private input exchange	Privileged information reference, excluded from fairness claims.

Table 2: Held-out mean entanglement surplus.

Method	Mean surplus
Classical exact	0.00000
Quantum static	0.22478
Quantum random	0.22685
Quantum greedy	0.30839
BER-MARL	**0.50642**
Online DP	0.57122
Future-clairvoyant	0.58326

increase was 64.2%. Random and static allocation were nearly identical, showing that access to Bell pairs alone was insufficient. Greedy improved substantially over them, while BER-MARL added a further long-horizon gain.

BER-MARL recovered 88.66% of online-DP surplus. Its one-sided 95% lower bound was 87.68%, below the registered 90% requirement. Four of five gates passed; the composite gate failed solely on this near-oracle condition. No budget or matching violation occurred.

The twelve held-out request types had Bell-minus-classical gaps from zero to 0.03830, with mean 0.01760. Retaining zero-gap requests prevents the benchmark from rewarding indiscriminate entanglement use.

7.1 Interpretation

The ordering random/static < greedy < BER-MARL < DP supports a specific mechanism. Immediate certified value matters, but future opportunity value and budget pacing add substantial reward. The remaining DP gap indicates incomplete zero-shot transfer to the new transition catalog. DP knows that catalog's exact transition matrix; BER-MARL receives no final-catalog adaptation period.

The classical-exact row defines surplus and intentionally uses no Bell pairs. It should not be interpreted as an equal-resource allocator baseline. Fair allocation comparisons are among random, static, greedy, BER-MARL, and DP.

8 Audit, Reproducibility, and Resources

An independent release program re-enumerated every held-out classical optimum, independently resolved every SDP ceiling, rebuilt all method means and bootstrap intervals from 23,000 raw episode rows, hashed all 16 checkpoints, and validated the authoritative bundles. Every reported scalar reproduced within 10^{-12}.

The confirmatory run retained 16 checkpoints, 16,000 learned-policy episode records, all baseline records, request certificates, resolved configurations, resource counters, and failures. It consumed 600,000 allocator decisions, 552,592 optimizer updates including request certification, 1,382,400 analytic circuit-equivalent evaluations, zero physical shots, and 956.91 wall-seconds on CPU. A fresh remote clone validated 281 reportable bundles, and the repository test suite contained 128 passing tests.

9 Limitations and Publication Scope

The outputs are abstract coordination bits, not actions in a deployed network or robot team. This abstraction enables exact certification but limits application validity. Bell correlations are ideal and analytic; the confirmatory study does not include finite shots, hardware noise, or entanglement-generation failures. The graph has four agents and a small enumerated matching action set. PPO, recurrent, threshold, and learned-model allocators were not locked primary baselines. Finally, the novelty screen was bounded and does not justify an unqualified "first" claim.

The current evidence supports a preprint and workshop submission. A selective conference or journal version should add: (i) systematic related work; (ii) application-grounded action semantics; (iii) finite-shot and noisy-resource experiments; (iv) PPO, model-based, recurrent, and threshold controls; (v) scaling over agents, horizons, and budgets; and (vi) a prospectively frozen online-adaptation study addressing the observed DP gap. A small hardware demonstration would further strengthen the quantum-systems claim.

10 Conclusion

We introduced a certified sequential benchmark for allocating consumable entanglement across a changing multi-agent graph and proposed BER-MARL, a masked DQN resource controller using only public pre-context information. On held-out request types, dynamics, and sequences, BER-MARL increased entanglement surplus by 64.2% over myopic greedy, with every one of 16 seeds positive and a paired 95% interval excluding zero. Exact classical enumeration, independent SDP bounds, paired evaluation, retained failures, and a separate audit support this conclusion.

The claim has a clear boundary. BER-MARL recovered 88.66% of model-aware DP value, but the preregistered one-sided lower bound did not exceed 90%; we reject a near-oracle transfer claim. The contribution is not a new proof of Bell nonlocality. It is the formulation, certification, and empirical solution of a scarce-entanglement routing problem. The strongest next step is to give the abstract actions application semantics and combine public-model adaptation with realistic noisy quantum resources.

References

[1] John S. Bell. On the einstein podolsky rosen paradox. *Physics Physique Fizika*, 1(3):195–200, 1964.

[2] John F. Clauser, Michael A. Horne, Abner Shimony, and Richard A. Holt. Proposed experiment to test local hidden-variable theories. *Physical Review Letters*, 23(15):880–884, 1969.

[3] Richard Cleve, Peter Hoyer, Ben Toner, and John Watrous. Consequences and limits of nonlocal strategies. In *Proceedings of the 19th IEEE Conference on Computational Complexity*, pages 236–249, 2004.

[4] Alexander DeRieux and Walid Saad. eqmarl: Entangled quantum multi-agent reinforcement learning for distributed cooperation over quantum channels. In *The Thirteenth International Conference on Learning Representations*, 2025.

[5] Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A. Rusu, Joel Veness, Marc G. Bellemare, Alex Graves, Martin Riedmiller, Andreas K. Fidjeland, Georg Ostrovski, Stig Petersen, Charles Beattie, Amir Sadik, Ioannis Antonoglou, Helen King, Dharshan Kumaran, Daan Wierstra, Shane Legg, and Demis Hassabis. Human-level control through deep reinforcement learning. *Nature*, 518:529–533, 2015.

[6] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

[7] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2 edition, 2018.

[8] Boris S. Tsirelson. Quantum generalizations of bell's inequality. *Letters in Mathematical Physics*, 4:93–100, 1980.